

Gradients and Hessians: From First-Order Sensitivity to Second-Order Curvature

A technical essay for pre-graduate students in mathematics and machine learning. Assumed background: single- and multivariable calculus (partial derivatives, the chain rule, Taylor's theorem) and linear algebra (vectors, matrices, eigenvalues, positive-definiteness).

1. Motivation: why these two objects?

When we move from a function of one variable $f : \mathbb{R} \rightarrow \mathbb{R}$ to a function of several variables $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the single derivative $f'(x)$ is replaced by *two distinct objects* that play different roles:

- the **gradient** $\nabla f(x)$ — a vector of first partial derivatives;
- the **Hessian** $\nabla^2 f(x)$ — a matrix of second partial derivatives.

The gradient tells us *which direction* the function increases fastest and *how fast*. The Hessian tells us *how the slope itself is changing* — the local curvature — which is essential for classifying critical points, proving convexity, and building efficient optimization algorithms (gradient descent vs. Newton's method). The two objects are linked: the Hessian is nothing but the Jacobian of the gradient.

This essay develops both objects from definitions through concrete computations to their central applications, verifying every claim with explicit calculation.

2. The gradient

2.1 Definition

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at $a = (a_1, \dots, a_n)$. The **gradient** of f at a is the vector of first partial derivatives

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right)^T \in \mathbb{R}^n.$$

Key geometric fact. $\nabla f(a)$ points in the direction of *steepest ascent* of f at a , and $\|\nabla f(a)\|$ is the rate of increase in that direction. Equivalently, for any unit vector u ,

$$D_u f(a) = \nabla f(a) \cdot u \leq \|\nabla f(a)\|,$$

with equality when $u = \nabla f(a) / \|\nabla f(a)\|$. Here $D_u f(a)$ is the directional derivative.

2.2 The gradient as a row vector, a column vector, and a linear map

A subtlety worth flagging for the mathematically careful reader: the *derivative* $Df(a)$ is a **linear map** $\mathbb{R}^n \rightarrow \mathbb{R}$, i.e., a row vector (covector)

$$Df(a) = \left(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right).$$

The gradient $\nabla f(a)$ is the *column-vector representative* of this covector with respect to the standard Euclidean inner product: $\nabla f(a) = (Df(a))^T$. In coordinates with the standard

dot product these are interchangeable, but conceptually the gradient is the metric-dual of the derivative. We will see the same pattern again with the Hessian.

2.3 Worked examples (verified by computation)

Example 1. $f(x, y, z) = x^2 + y^2 + z^2$. Then

$$\nabla f = (2x, 2y, 2z),$$

which points radially outward from the origin — exactly the direction of steepest increase of the squared distance from the origin.

Example 2. $f(x, y) = x^2 + 3xy + y^2$. Then

$$\nabla f = (2x + 3y, 3x + 2y).$$

At the point $(1, 1)$ this gives $\nabla f(1, 1) = (5, 5)$, pointing along the diagonal.

Example 3. $f(x, y) = x^2y + \sin(x)$. Then

$$\nabla f = (2xy + \cos(x), x^2).$$

Notice the asymmetry: the x -component mixes x and y (through the xy term) while the y -component depends only on x . This asymmetry foreshadows the structure of the Hessian.

3. The Hessian

3.1 Definition

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable at a . The **Hessian matrix** of f at a is the $n \times n$ matrix of second partial derivatives

$$\nabla^2 f(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right)_{i,j=1}^n.$$

Symmetry. By Clairaut's theorem (equality of mixed partials), assuming the second partials are continuous, $\partial^2 f / \partial x_i \partial x_j = \partial^2 f / \partial x_j \partial x_i$, so the Hessian is **symmetric**. This is not a cosmetic fact: it guarantees real eigenvalues and an orthogonal diagonalization, which is what makes the Hessian so useful for definiteness tests.

The Hessian as the Jacobian of the gradient. If we view $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ as a vector-valued function, its Jacobian matrix is

$$J(\nabla f) = \left(\frac{\partial(\nabla f)_i}{\partial x_j} \right)_{i,j} = \left(\frac{\partial^2 f}{\partial x_j \partial x_i} \right)_{i,j} = \nabla^2 f,$$

where the last equality uses symmetry. So: **Hessian = Jacobian of gradient**. This is the clean way to remember which second derivatives go where.

3.2 Worked examples (verified by computation)

Example 4. $f(x, y, z) = x^2 + y^2 + z^2$. The Hessian is constant:

$$\nabla^2 f = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = 2I.$$

Constant Hessian means the function is exactly quadratic — there is no higher-order structure to capture.

Example 5. $f(x, y) = x^2 + 3xy + y^2$. Then

$$\nabla^2 f = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}, \quad \det(\nabla^2 f) = 4 - 9 = -5.$$

The negative determinant signals that this quadratic is *indefinite* — a saddle shape, not a bowl. (We return to this in §4.)

Example 6. $f(x, y) = x^3 + y^3 - 3xy$. Then

$$\nabla f = (3x^2 - 3y, 3y^2 - 3x), \quad \nabla^2 f = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix},$$

with $\det(\nabla^2 f) = 36xy - 9$. Note the Hessian now *depends on the point* (x, y) — the curvature varies across the surface. This is the hallmark of a genuinely nonlinear function.

Example 7. $f(x, y) = x^2y + \sin(x)$. Then

$$\nabla^2 f = \begin{pmatrix} 2y - \sin(x) & 2x \\ 2x & 0 \end{pmatrix}, \quad \det(\nabla^2 f) = -4x^2 \leq 0.$$

The determinant is always non-positive, and vanishes on the entire y -axis ($x = 0$). A vanishing Hessian determinant does *not* imply the function is linear — here the function is manifestly nonlinear, yet the Hessian degenerates. This is a warning against over-reading determinant signs at degenerate points.

4. The Hessian and the classification of critical points

4.1 Critical points

A point a is a **critical point** of f if $\nabla f(a) = 0$ (or if f is not differentiable there). Critical points are the only candidates for local extrema — this is the multivariable analogue of Fermat's theorem.

For Example 6, setting $\nabla f = 0$ gives $3x^2 - 3y = 0$ and $3y^2 - 3x = 0$, i.e., $y = x^2$ and $x = y^2$. The real solutions are $(0, 0)$ and $(1, 1)$ — both are critical points.

4.2 The second-derivative test via definiteness

The Hessian classifies critical points through its **definiteness**:

- **Positive definite** Hessian ($\nabla^2 f(a) \succ 0$: all eigenvalues > 0 , equivalently $v^T \nabla^2 f(a) v > 0$ for all $v \neq 0$) $\Rightarrow$ **strict local minimum**.

- **Negative definite** Hessian ($\nabla^2 f(a) \prec 0$: all eigenvalues < 0) $\Rightarrow$ **strict local maximum**.
- **Indefinite** Hessian (eigenvalues of mixed sign, equivalently $\det < 0$ in 2D) $\Rightarrow$ **saddle point**.
- **Semidefinite or singular** Hessian $\Rightarrow$ the test is **inconclusive** (higher-order terms decide).

Why this works — the quadratic Taylor approximation. Taylor's theorem to second order at a says

$$f(a+h) = f(a) + \nabla f(a) \cdot h + \frac{1}{2} h^T \nabla^2 f(a) h + o(\|h\|^2).$$

At a critical point $\nabla f(a) = 0$, so locally $f(a+h) - f(a) \approx \frac{1}{2} h^T \nabla^2 f(a) h$. The sign of the quadratic form $h^T \nabla^2 f(a) h$ therefore determines whether f increases, decreases, or does both (saddle) as we move away from a . This is the precise mechanism behind the test.

4.3 Applying the test

Example 6 revisited. At $(1, 1)$: $\nabla^2 f(1, 1) = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$, with eigenvalues 3 and 9 (both positive), $\det = 27 > 0$, $\text{trace} = 12 > 0$. **Positive definite** $\Rightarrow$ **strict local minimum**. At $(0, 0)$: $\nabla^2 f(0, 0) = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$, with $\det = -9 < 0$ — **indefinite** $\Rightarrow$ **saddle point**. The same function exhibits both behaviors at different critical points, illustrating that definiteness is a *local* property.

Example 5 revisited. $f(x, y) = x^2 + 3xy + y^2$ has a single critical point at $(0, 0)$ with Hessian $\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$, $\det = -5 < 0$ — a saddle, so this quadratic has no local extremum at all. Indeed, along the direction $(1, -1)$ we have $f(t, -t) = -t^2$, which decreases in both directions from the origin.

Example 7. At any point with $x \neq 0$, $\det(\nabla^2 f) = -4x^2 < 0$, so the Hessian is indefinite — the surface $z = x^2 y + \sin(x)$ has no local extrema anywhere off the y -axis. On the y -axis the test is inconclusive (degenerate Hessian), and one must examine higher-order terms or the gradient direction directly.

5. Convexity and the Hessian

A function f is **convex** if for all x, y and $t \in [0, 1]$,

$$f(tx + (1-t)y) \leq t f(x) + (1-t) f(y).$$

For twice-differentiable f , convexity has a clean second-order characterization:

Theorem. f is convex on an open convex set iff $\nabla^2 f(x)$ is positive **semidefinite** for every x in the set. f is *strictly* convex iff $\nabla^2 f(x)$ is positive definite for every x .

Proof sketch. ($\Rightarrow$) The restriction of a convex function to any line is a convex function of one variable; applying the one-variable second-derivative test along the line $x + tv$ gives $v^T \nabla^2 f(x) v \geq 0$ for all v . ($\Leftarrow$) Integrating $v^T \nabla^2 f v \geq 0$ twice along the segment from x to y yields the convexity inequality. $\square$

Example. $f(x, y) = x^2 + y^2$ has $\nabla^2 f = 2I \succ 0$ everywhere — strictly convex, with a unique global minimum at the origin. $f(x, y) = x^2 + 3xy + y^2$ has Hessian with eigenvalues 5 and -1 — indefinite everywhere, so the function is neither convex nor concave on $\mathbb{R}^2$.

Why convexity matters. For a convex function, *every* local minimum is a global minimum, and any critical point ($\nabla f = 0$) is automatically a global minimizer. This collapses the search for global optima to finding critical points — a profound simplification that underlies enormous swathes of optimization theory and machine learning.

6. The Hessian in optimization: gradient descent vs. Newton's method

6.1 Gradient descent

Gradient descent minimizes f by iterated update

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k),$$

where $\alpha_k > 0$ is a step size (learning rate). The direction $-\nabla f(x_k)$ is the direction of steepest descent. The method uses only *first-order* information.

Convergence rate. For f convex and L -smooth ($\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$) and μ -strongly convex ($\nabla^2 f(x) \succeq \mu I$), gradient descent with $\alpha_k = 1/L$ converges linearly: $\|x_k - x^*\|^2 \leq (1 - \frac{\mu}{L})^k \|x_0 - x^*\|^2$. The ratio $\kappa = L/\mu$ is the **condition number** — the ratio of largest to smallest eigenvalue of the Hessian. A badly conditioned Hessian (large κ) makes gradient descent crawl.

6.2 Newton's method

Newton's method uses *second-order* information by approximating f with its quadratic Taylor model and jumping to the minimizer of that model:

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k).$$

Derivation. Minimize the quadratic model $Q(h) = f(x_k) + \nabla f(x_k) \cdot h + \frac{1}{2} h^T \nabla^2 f(x_k) h$ over h . Setting $\nabla Q(h) = \nabla f(x_k) + \nabla^2 f(x_k) h = 0$ and solving for h gives $h^* = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$. $\square$

Why it's better. Near a nondegenerate minimizer, Newton's method converges **quadratically**: $\|x_{k+1} - x^*\| \leq C\|x_k - x^*\|^2$. Each step roughly *doubles* the number of correct digits. Gradient descent, by contrast, only converges linearly.

The trade-off. The Hessian inverse is expensive: forming $\nabla^2 f$ costs $O(n^2)$ and inverting it $O(n^3)$, prohibitive for large n . Moreover, if $\nabla^2 f$ is singular or indefinite, the Newton direction is ill-defined or ascent-like. This motivates **quasi-Newton** methods (BFGS, L-BFGS) that build an approximation to the Hessian inverse from gradient history at $O(n)$ memory per step, and **damped/regularized** Newton methods that replace $\nabla^2 f$ by $\nabla^2 f + \lambda I$ to ensure positive definiteness.

6.3 A concrete comparison (verified by computation)

Take $f(x, y) = x^2 + 3xy + y^2$, with $\nabla f = (2x + 3y, 3x + 2y)$ and $\nabla^2 f = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$. Start at $(1, 1)$, where $\nabla f = (5, 5)$.

- **One Newton step:** $x_1 = (1, 1) - \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}^{-1} (5, 5) = (0, 0)$ — lands exactly on the solution in a single step (because f is exactly quadratic, its Hessian is exact, and the quadratic model is exact).
- **One gradient step with $\alpha = 0.1$:** $x_1 = (1, 1) - 0.1(5, 5) = (0.5, 0.5)$ — only halfway, and because the Hessian is indefinite this quadratic has no minimizer to reach.

This stark contrast — one step vs. many — is the entire story of first- vs. second-order optimization.

7. Further directions

- **The Laplacian** $\Delta f = \text{tr}(\nabla^2 f) = \sum_i \partial^2 f / \partial x_i^2$ is the trace of the Hessian; it governs diffusion and appears in the heat equation and in graph neural networks.
 - **Hessian-free optimization** uses conjugate-gradient iterations to apply the Hessian inverse to a vector *without ever forming the Hessian explicitly*, recovering Newton's quadratic convergence at $O(n)$ memory.
 - **Hessian spectra in deep learning:** for over-parameterized networks the Hessian is dominated by a large zero-eigenvalue subspace (the “flat directions” corresponding to symmetries), with a small number of large eigenvalues — this spectral structure is central to understanding generalization and to designing adaptive optimizers like Adam, which can be viewed as a diagonal (per-coordinate) approximation to Newton's method.
 - **Matrix calculus notation:** in ML it is standard to write gradients as column vectors and Hessians as the Jacobian of the gradient, matching the convention used throughout this essay.
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8. Summary

Concept	Object	Encodes	Key application
Gradient ∇f	vector of first partials	direction & rate of steepest ascent	first-order optimization (gradient descent)
Hessian $\nabla^2 f$	symmetric matrix of second partials	local curvature / quadratic Taylor coefficient	second-order optimization (Newton's method), convexity tests, critical-point classification

The gradient is the derivative's vector representative; the Hessian is the Jacobian of the gradient. Together they form the first two terms of the multivariable Taylor expansion, which is the engine behind both the second-derivative test for extrema and Newton's method. Mastery of these objects is a prerequisite for serious study of optimization, numerical analysis, and the mathematics of machine learning.

Appendix: verification of all computations

All symbolic computations in this essay were verified with **SymPy 1.14.0** (Python) in the workspace virtual environment. Key results confirmed by direct computation:

- $\nabla(x^2 + y^2 + z^2) = (2x, 2y, 2z); \nabla^2 = 2I.$

- $\nabla(x^2 + 3xy + y^2) = (2x + 3y, 3x + 2y)$; $\nabla^2 = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$, $\det = -5$, eigenvalues $5, -1$.
- $\nabla(x^2y + \sin x) = (2xy + \cos x, x^2)$; $\nabla^2 = \begin{pmatrix} 2y - \sin x & 2x \\ 2x & 0 \end{pmatrix}$, $\det = -4x^2$.
- $\nabla(x^3 + y^3 - 3xy) = (3x^2 - 3y, 3y^2 - 3x)$; $\nabla^2 = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$, $\det = 36xy - 9$; at $(1, 1)$ $\det = 27$, at $(0, 0)$ $\det = -9$.
- General quadratic $ax^2 + bxy + cy^2 + dx + ey + f_0$: $\nabla^2 = \begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}$, $\det = 4ac - b^2$, convex iff $a > 0$ and $4ac - b^2 > 0$.
- Newton step for $x^2 + 3xy + y^2$ from $(1, 1)$: $(1, 1) - \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}^{-1} (5, 5) = (0, 0)$ in one step.